Table 4.

Taking the following values, we can explain the functioning of multiplier

MPC	MPS	K
0.00	1.00	1
0.10	0.90	1.11
0.50	0.50	2.00
0.75	0.25	4.00
0.90	0.10	10.00
1.00	0.00	α

C = 100 + 0.8 y; I = 100

I = 10

$$Y = C + I$$

Y = 100 + 0.8y + 100

1y − 0.8y = 100 + 100

0.2y = 200

Y = 1000

Here, C = 100 + 0.8y = 100 + (1000) = 900;

S = 100 = I

After I is raised by 10, now I = 110,

Y = 100 + 0.8y + 110

0.2y = 210

$$Y = \frac{210}{0.2} = 1050$$

Here C = 100=0.8(1050) = 940; S = 110 = I

Diagrammatic Explanation.

At 45° line y = C+ S

It implies the variables in X axis and Y axis are equal.

The MPC is assumed to be at 0.8 (C = 100 + 0.8y)

The aggregate demand (C+I) curve intersects 45° line at point E.

The original national income is 500.
(C = 100 + 0.8y = 100 + 0.8(500) = 500)

When I is 100, y = 1000, C = 900;
S = 100 = I

The new aggregate demand curve is C+I' = 100 + 0.8y + 100 +10

$$Y = \frac{210}{0.2} = 1050$$

C = 940; S = 110 = I

4.4.3. Working of Multiplier

Suppose the Government undertakes investment expenditure equal to ₹100 crore on some public works, by way of wages, price of materials etc. Thus income of labourers and suppliers of materials increases by ₹100 crore. Suppose the MPC is 0.8 that is 80 %. A sum of ₹80 crores is spent on consumption (A sum of ₹20 Crores is saved). As a result, suppliers of goods get an income of ₹80 crores. They inturn spend ₹64 crores (80% of ₹80 cr). In this manner consumption expenditure and increase in income act in a chain like manner.

Positive Multiplier and Negative Multiplier Effects

Positive Multiplier

When an initial increase in an injection of investment (or a decrease in a leakage) leads to a greater final increase in real GDP.

Negative Multiplier

When an initial increase in an injection of investment (or an increase in a leakage) leads to a greater final decrease in real GDP.

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